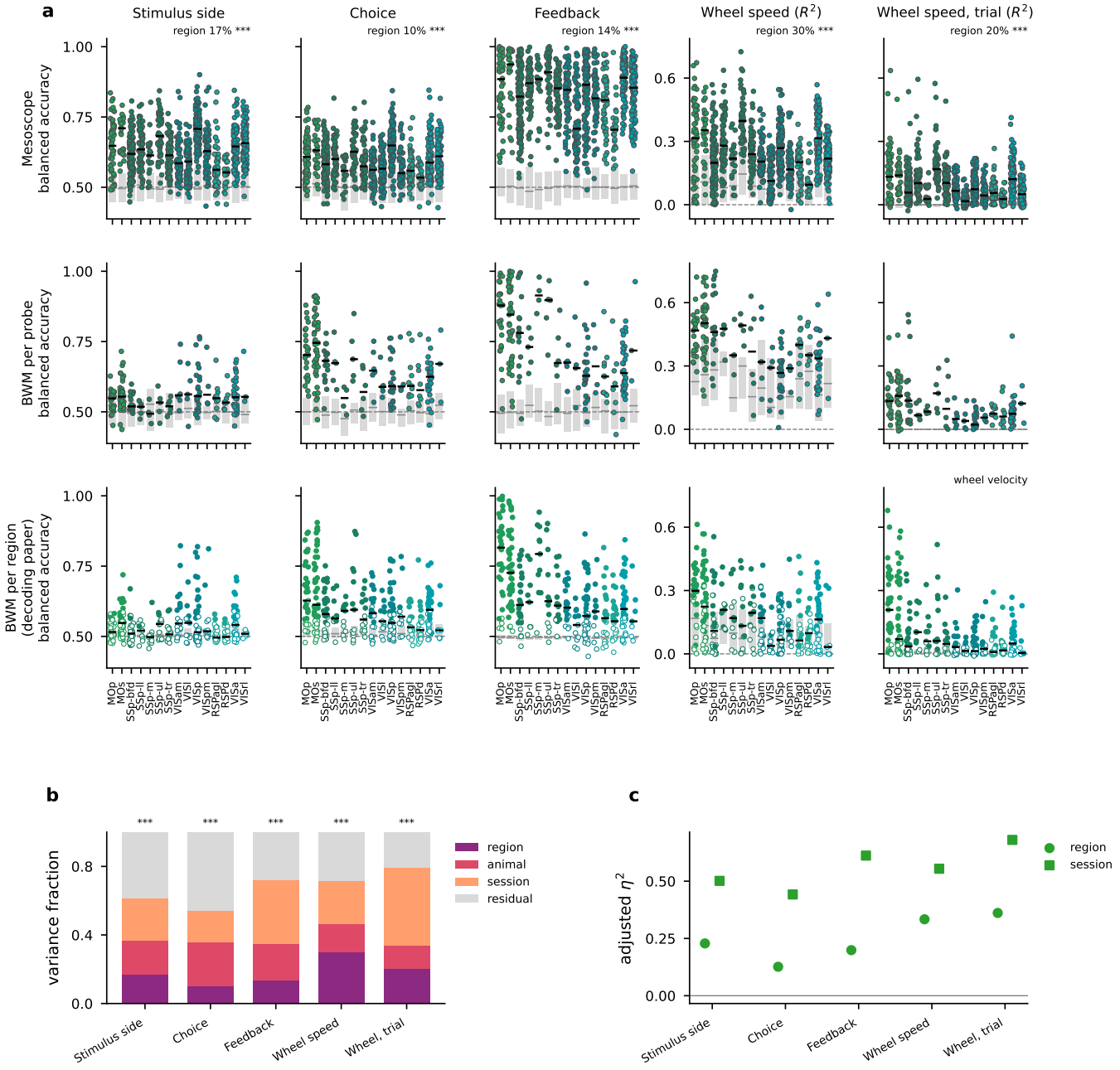


Figure 1: Variance of decoding across regions and sessions (all neurons). The 15 regions with ≥3 mesoscope sessions and BWM probes. Methods: *Region and session variance*. **a** Scores per region (columns ordered as in the Allen structure tree; Allen colours; bar: median; dashed: chance; grey band and line: 5–95th percentile and median of the baseline; for trial-specific wheel speed the shuffle baseline lies at $R^2 \approx 0$ and is hidden by the dashed line). Top: mesoscope, one dot per session and region from all its neurons there (149 sessions, 12 animals); baseline: shuffle control. Middle: BWM, one dot per probe from all its isocortex units, placed in the region where it enters the brain (198 probes, 93 animals; the majority region of its superficial units for 66% of probes); baseline: shuffle control. Bottom: BWM decoding paper, one dot per session and region from the region's units (346 pairs, 263 sessions, 116 animals); filled: $p < 0.05$ against pseudo-sessions (21–70% of dots); baseline: median of the pseudo-session null; wheel velocity in place of trial-specific wheel speed. Y axes shared within balanced accuracy and within R^2. Inset, top row: region share of variance from **b**. **b** Mesoscope variance fractions from a mixed model, score ∼ log N + (1|region) + (1|animal) + (1|session); stars: likelihood-ratio test of the region component. Region: stimulus side 17%, choice 10%, feedback 14%, wheel speed 30%, wheel, trial 20%; session plus animal: stimulus side 45%, choice 44%, feedback 58%, wheel speed 41%, wheel, trial 59%. **c** The same question as **b**, do scores differ more between regions or between sessions, but without a statistical model, using only reshuffling: adjusted η^2 of region (circles, within sessions) and of session (squares, within regions), 0 = chance. Filled: $p < 0.05$.